Design of Experiments and Statistical Techniques (TXL783)

Lecture 3

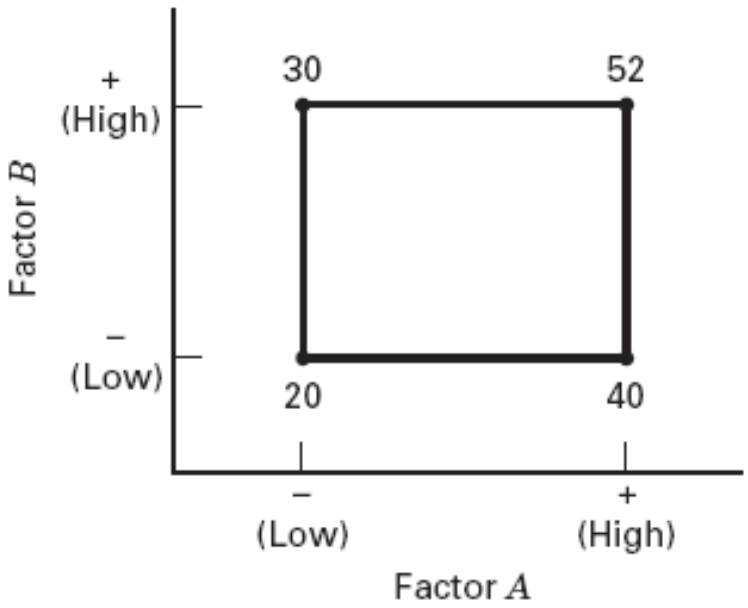
Factorial Design : Full factorial design

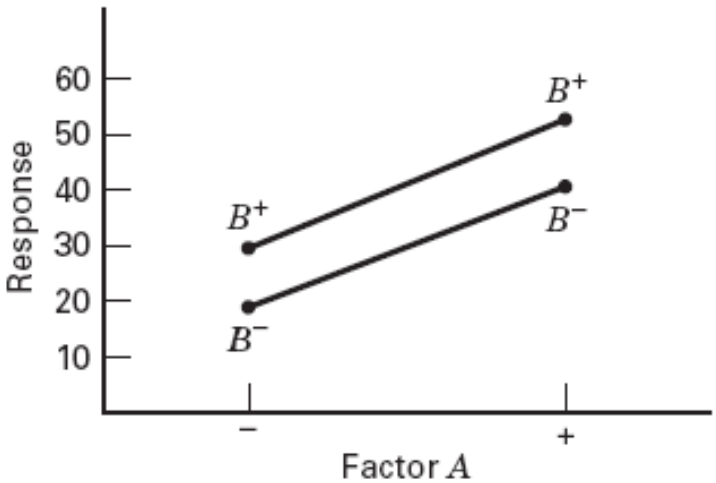
- A full factorial design of experiments consists of the following:
- *Vary **one factor** at a time*
- *Perform experiments for **all levels** of all factors*
- *Hence perform a large number of experiments that are needed!*
- *All **interactions** are captured.*
- Consider a simple design for the following case:
- Let the number of factors = k
- Let the number of levels for the ith factor = n_i
- The total number of experiments (N) that need to be performed is…..

Key Points of Factorial Designs

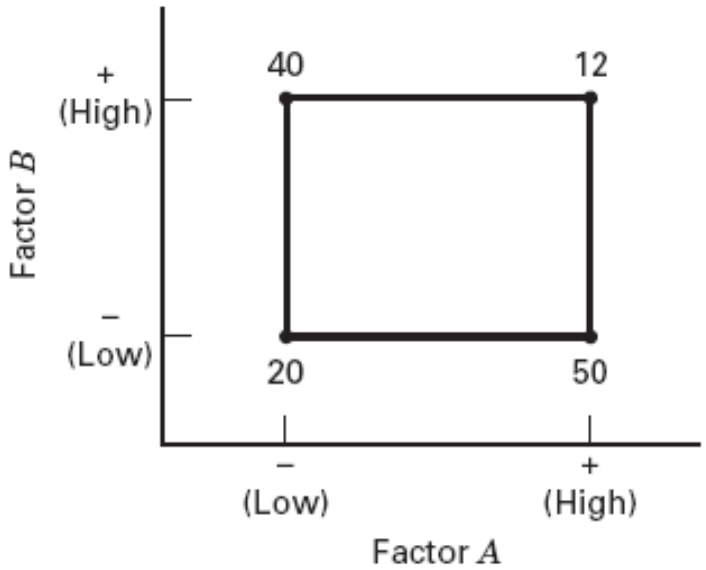
- If there are **a** levels of factor A and **b** levels of factor B, each replicate contains all **ab** treatment combinations. When factors are arranged in a factorial design, they are often said to be **crossed**.
- The effect of a factor is defined to be the change in response produced by a change in the level of the factor. This is frequently called a **main effect.**

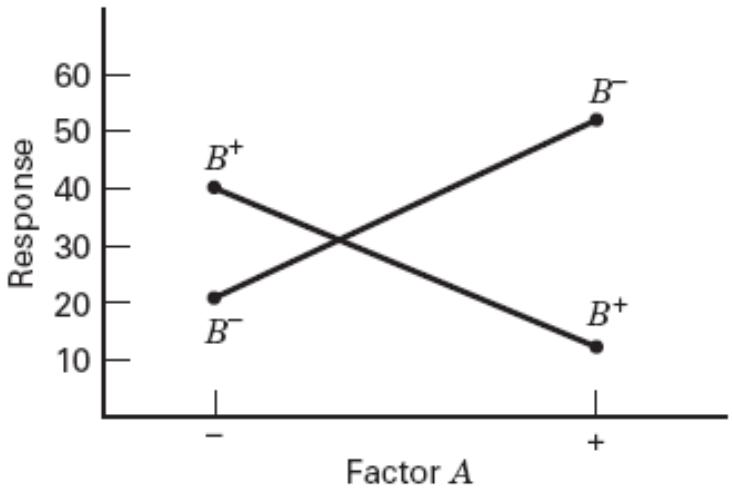
2- Factor Factorial Experiment



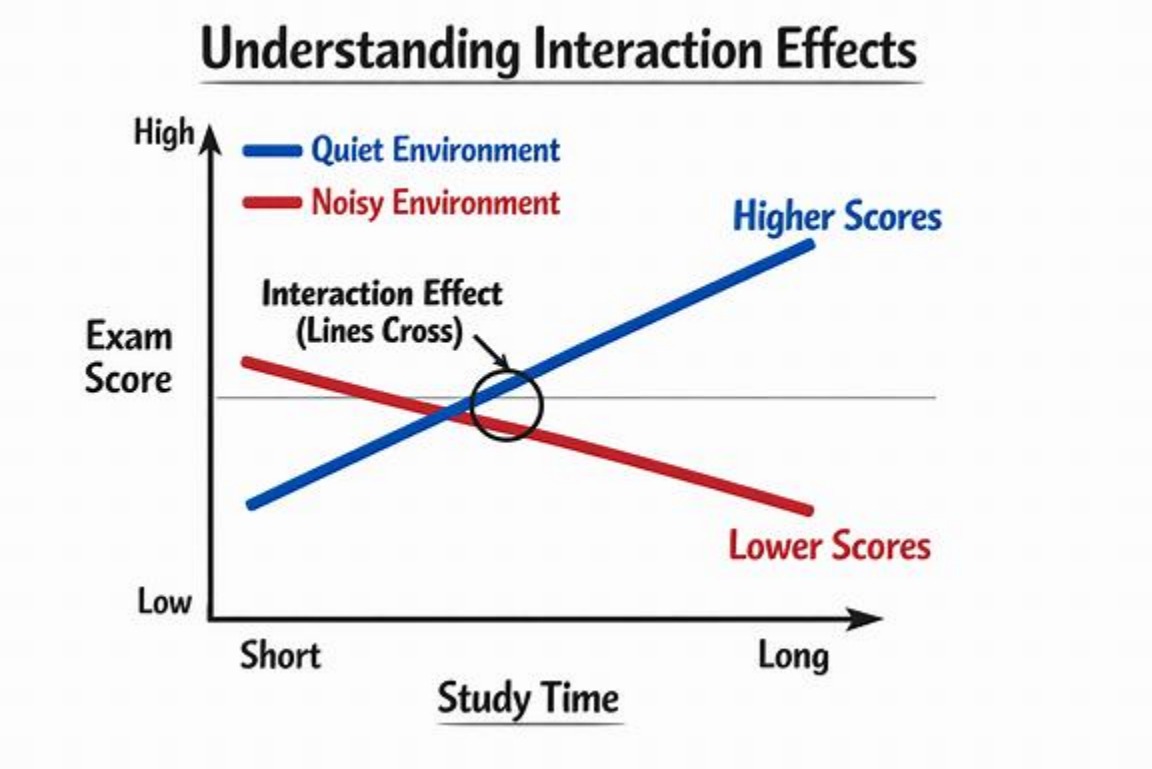


2- Factor Factorial Experiment with Interaction





Interaction Effect



Quantifying the Concept of Interaction

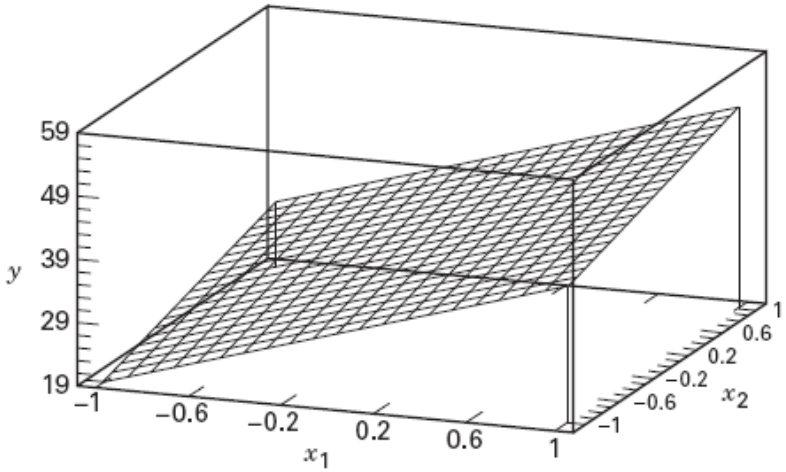
Regression Equation

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \epsilon$$

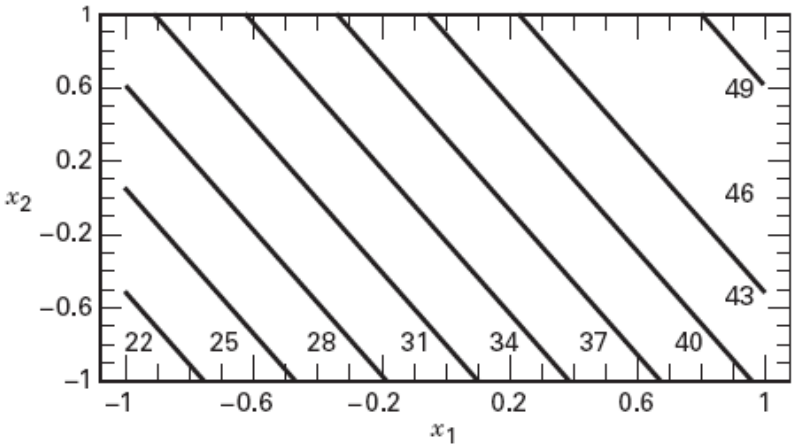
where y is the response, the β's are parameters whose values are to be determined, x_1 is a variable that represents factor A, x_2 is a variable that represents factor B, and ϵ is a random error term. The variables x_1 and x_2 are defined on a **coded scale** from -1 to $+1$ (the low and high levels of A and B), and $x_1 x_2$ represents the interaction between x_1 and x_2.

In a two-level factorial experiment using coded levels –1 and +1, why are the regression coefficients (β-values) equal to half the effect size of the corresponding factor or interaction?

Plots for the Model

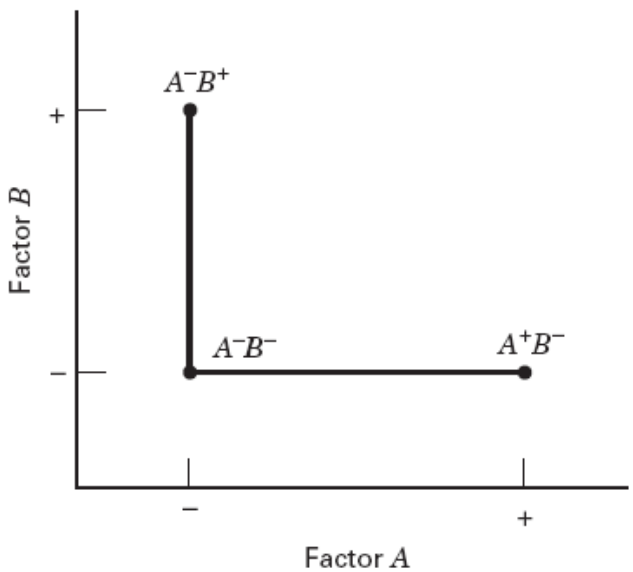


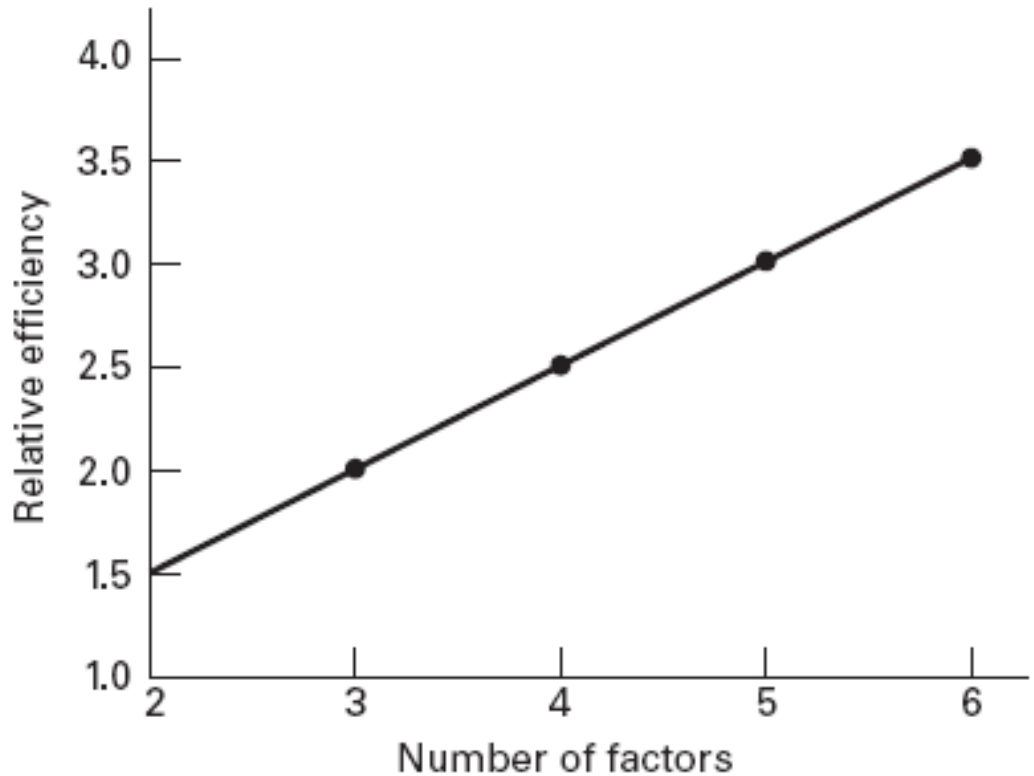
(a) The response surface



(b) The contour plot

Relative Efficiency of Factorial Design



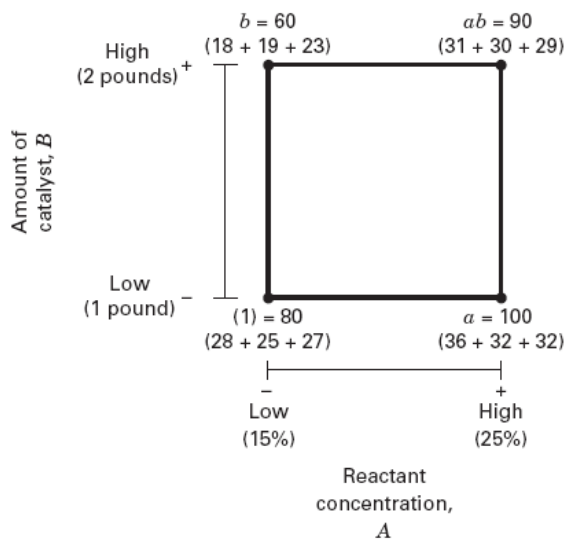


Advantages of Full factorial design

- More efficient that one-factor-at-a-time (OFAT) experiments.
- Factorial designs are necessary when interactions may be present to avoid misleading conclusions.
- Factorials design allow the effects of a factor to be estimated at several levels of the other factors, yielding conclusions that are valid under a range of experimental conditions.

2^2 Factorial Design

Factor A	Factor B	Treatment Combination	Replicate I	Replicate II	Replicate III	Total
−	−	*A* low, *B* low	28	25	27	80
+	−	*A* high, *B* low	36	32	32	100
−	+	*A* low, *B* high	18	19	23	60
+	+	*A* high, *B* high	31	30	29	90



Main and Interaction Effects

$$A = \bar{y}_{A^+} - \bar{y}_{A^-}$$
$$= \frac{ab + a}{2n} - \frac{b + (1)}{2n}$$
$$= \frac{1}{2n}[ab + a - b - (1)]$$

Contrast

$$B = \bar{y}_{B^+} - \bar{y}_{B^-}$$
$$= \frac{ab + b}{2n} - \frac{a + (1)}{2n}$$
$$= \frac{1}{2n}[ab + b - a - (1)]$$

$$AB = \frac{ab + (1)}{2n} - \frac{a + b}{2n}$$
$$= \frac{1}{2n}[ab + (1) - a - b]$$

Sum of the Squares

$$SS_A = \frac{[ab + a - b - (1)]^2}{4n}$$

$$SS_B = \frac{[ab + b - a - (1)]^2}{4n}$$

$$SS_{AB} = \frac{[ab + (1) - a - b]^2}{4n}$$

$$\boxed{SS_T = \sum y_i^2 - \frac{(\sum y_i)^2}{N}}$$

ANOVA

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_q
A	208.33	1	208.33	53.15
B	75.00	1	75.00	19.13
AB	8.33	1	8.33	2.13
Error	31.34	8	3.92	
Total	323.00	11		

$$x_1 = \frac{\text{Conc} - (\text{Conc}_{\text{low}} + \text{Conc}_{\text{high}})/2}{(\text{Conc}_{\text{high}} - \text{Conc}_{\text{low}})/2}$$

$$x_2 = \frac{\text{Catalyst} - (\text{Catalyst}_{\text{low}} + \text{Catalyst}_{\text{high}})/2}{(\text{Catalyst}_{\text{high}} - \text{Catalyst}_{\text{low}})/2}$$

Coded and Natural Variables

$$x_1 = \frac{\text{Conc} - (\text{Conc}_{\text{low}} + \text{Conc}_{\text{high}})/2}{(\text{Conc}_{\text{high}} - \text{Conc}_{\text{low}})/2}$$

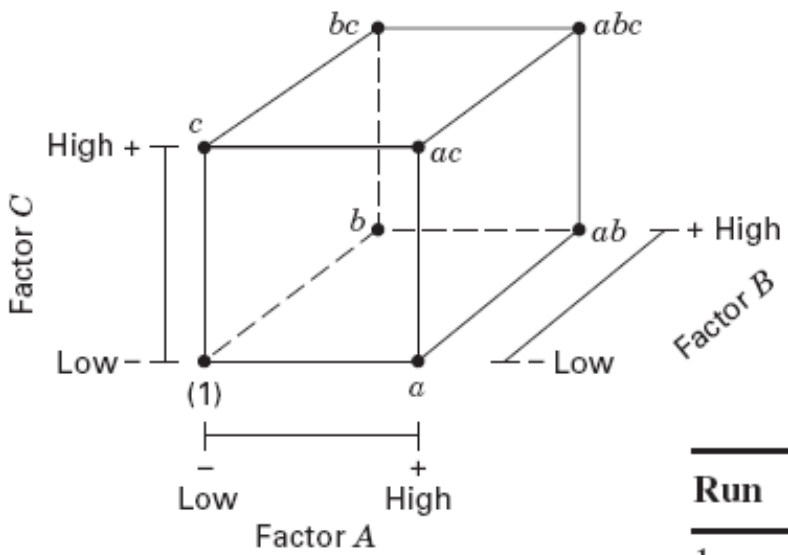
$$x_2 = \frac{\text{Catalyst} - (\text{Catalyst}_{\text{low}} + \text{Catalyst}_{\text{high}})/2}{(\text{Catalyst}_{\text{high}} - \text{Catalyst}_{\text{low}})/2}$$

$$x_1 = \frac{\text{Conc} - (15 + 25)/2}{(25 - 15)/2} = \frac{\text{Conc} - 20}{5}$$

$$x_2 = \frac{\text{Catalyst} - (1 + 2)/2}{(2 - 1)/2} = \frac{\text{Catalyst} - 1.5}{0.5}$$

$$\hat{y} = 27.5 + \left(\frac{8.33}{2}\right) x_1 + \left(\frac{-5.00}{2}\right) x_2$$

2^3 Factorial Design



Design Matrix

Run	A	B	C	Labels
1	−	−	−	(1)
2	+	−	−	a
3	−	+	−	b
4	+	+	−	ab
5	−	−	+	c
6	+	−	+	ac
7	−	+	+	bc
8	+	+	+	abc

Sign Table

For proper factorial design:

$$\sum A = 0$$

$$\sum B = 0$$

$$\sum AB = 0$$

and AB are

and more importantly

$$\sum AB \cdot A = 0$$

$$\sum AB \cdot B = 0$$

$$\sum A \cdot B = 0$$

Also

Orthogonality in DOE means that the algebraic sum of the products of coefficients of any two factor contrasts is zero, ensuring independent estimation of factorial effects.

Main effects

$$A = \frac{1}{4n}[a - (1) + ab - b + ac - c + abc - bc]$$

$$A = \bar{y}_{A^+} - \bar{y}_{A^-}$$

$$= \frac{a + ab + ac + abc}{4n} - \frac{(1) + b + c + bc}{4n}$$

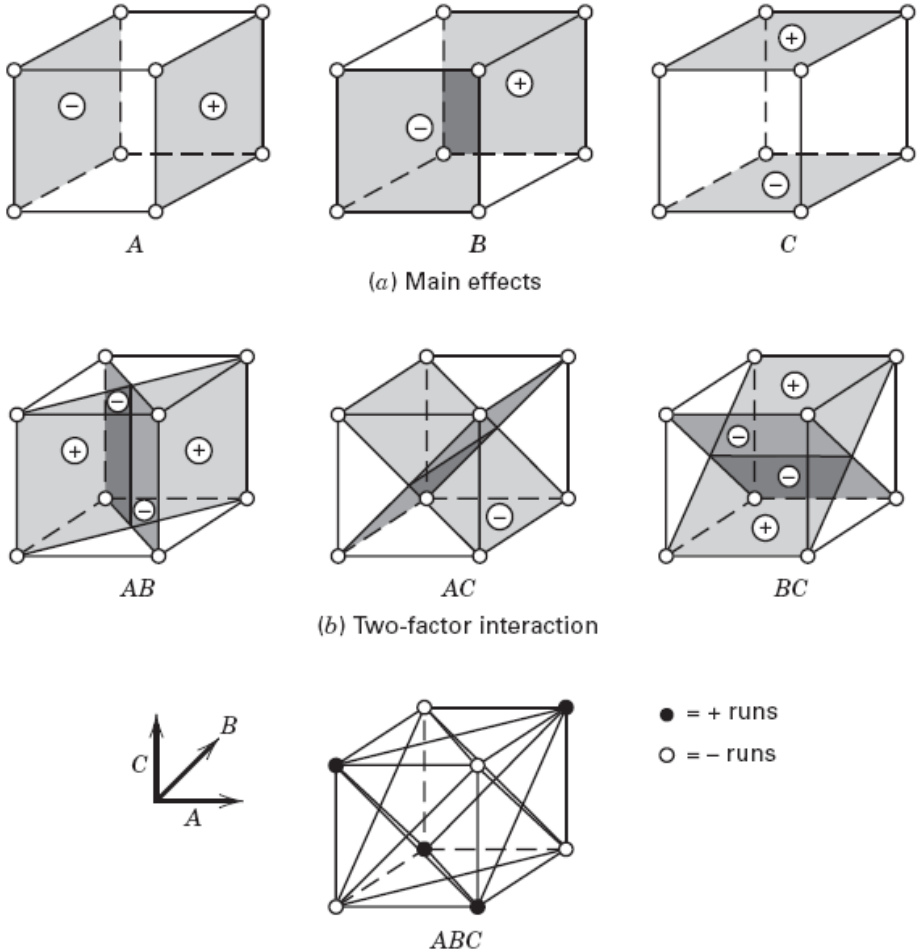
$$B = \bar{y}_{B^+} - \bar{y}_{B^-}$$

$$= \frac{1}{4n}[b + ab + bc + abc - (1) - a - c - ac]$$

$$C = \bar{y}_{C^+} - \bar{y}_{C^-}$$

$$= \frac{1}{4n}[c + ac + bc + abc - (1) - a - b - ab]$$

Three-factor Interaction



Interaction effects

B	Average *A* Effect
High (+)	$\frac{[(abc - bc) + (ab - b)]}{2n}$
Low (−)	$\frac{\{(ac - c) + [a - (1)]\}}{2n}$
Difference	$\frac{[abc - bc + ab - b - ac + c - a + (1)]}{2n}$

$$AB = \frac{[abc - bc + ab - b - ac + c - a + (1)]}{4n}$$

$$AB = \frac{abc + ab + c + (1)}{4n} - \frac{bc + b + ac + a}{4n}$$

Interaction effects (Cont.)

$$AC = \frac{1}{4n}[(1) - a + b - ab - c + ac - bc + abc]$$

$$BC = \frac{1}{4n}[(1) + a - b - ab - c - ac + bc + abc]$$

$$ABC = \frac{1}{4n}\{[abc - bc] - [ac - c] - [ab - b] + [a - (1)]\}$$

$$= \frac{1}{4n}[abc - bc - ac + c - ab + b + a - (1)]$$

Calculating Effects in 2^3 design

Algebraic Signs for Calculating Effects in the 2^3 Design

Treatment Combination	Factorial Effect							
	I	A	B	AB	C	AC	BC	ABC
(1)	+	−	−	+	−	+	+	−
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
ab	+	+	+	+	−	−	−	−
c	+	−	−	+	+	−	−	+
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
abc	+	+	+	+	+	+	+	+

The I column consists entirely of plus signs (+). This is because it is used to estimate the **grand average** of all observations in the experiment. When you multiply the response values by the signs in this column and divide by the total number of runs, you are essentially calculating the mean of the entire data set.

Example

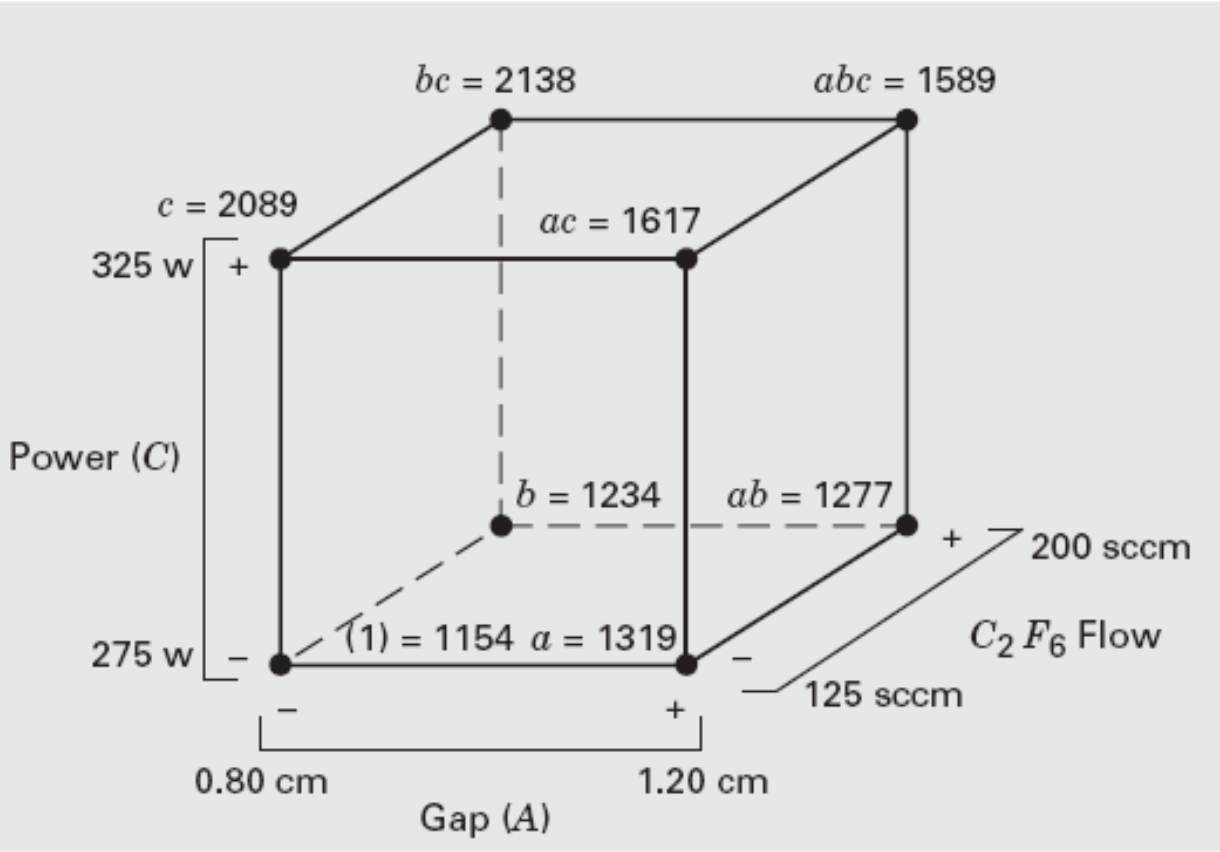
Run	Coded Factors			Etch Rate	
	A	*B*	*C*	Replicate 1	Replicate 2
1	−1	−1	−1	550	604
2	1	−1	−1	669	650
3	−1	1	−1	633	601
4	1	1	−1	642	635
5	−1	−1	1	1037	1052
6	1	−1	1	749	868
7	−1	1	1	1075	1063
8	1	1	1	729	860

Run	Coded Factors A	B	C	Etch Rate Replicate 1	Replicate 2	Total
1	−1	−1	−1	550	604	(1) = 1154
2	1	−1	−1	669	650	a = 1319
3	−1	1	−1	633	601	b = 1234
4	1	1	−1	642	635	ab = 1277
5	−1	−1	1	1037	1052	c = 2089
6	1	−1	1	749	868	ac = 1617
7	−1	1	1	1075	1063	bc = 2138
8	1	1	1	729	860	abc = 1589

Factor Levels	Low (−1)	High (+1)
A (Gap, cm)	0.80	1.20
B (C_2F_6 flow, SCCM)	125	200
C (Power, W)	275	325

- Estimate the factor effects. Which effects appear to be large?
- Formulate the ANOVA
- Write down the natural and coded regression model for predicting the etch rate.
- Analyze the coefficient of determination, the contribution of factors and residuals.

Example


bc = 2138
abc = 1589
c = 2089
ac = 1617
325 w
+
Power (C)
b = 1234
ab = 1277
+
200 sccm
(1) = 1154
a = 1319
275 w
–
–
C2 F6 Flow
125 sccm
–
+
0.80 cm
1.20 cm
Gap (A)

Estimate Summary

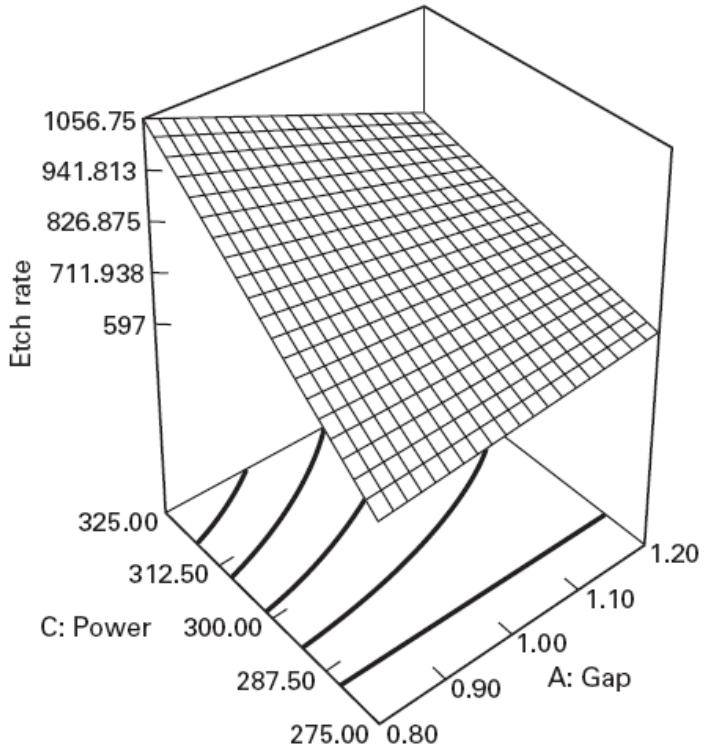
Factor	Effect Estimate	Sum of Squares	Percent Contribution
A	−101.625	41,310.5625	7.7736
B	7.375	217.5625	0.0409
C	306.125	374,850.0625	70.5373
AB	−24.875	2475.0625	0.4657
AC	−153.625	94,402.5625	17.7642
BC	−2.125	18.0625	0.0034
ABC	5.625	126.5625	0.0238

ANOVA

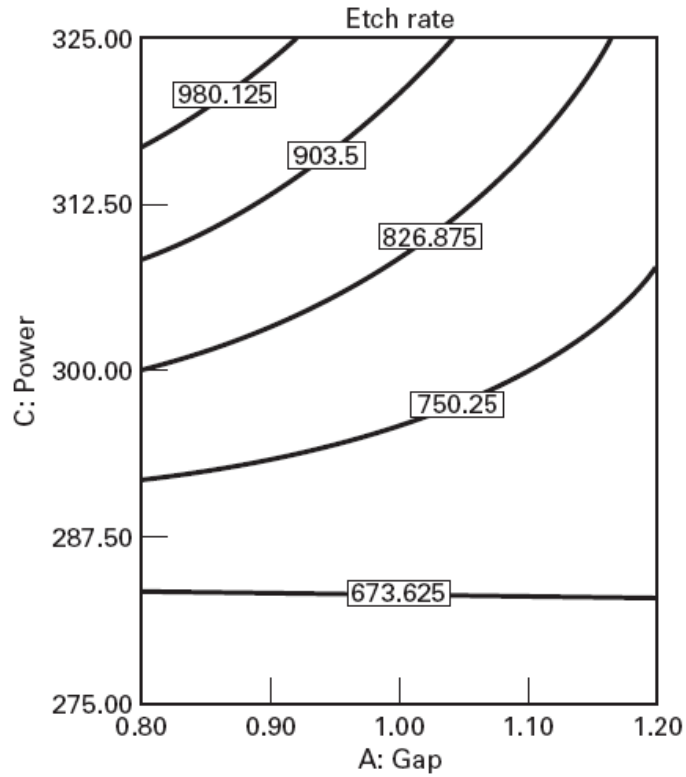
Analysis of Variance for the Plasma Etching Experiment

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P-Value
Gap (A)	41,310.5625	1	41,310.5625	18.34	0.0027
Gas flow (B)	217.5625	1	217.5625	0.10	0.7639
Power (C)	374,850.0625	1	374,850.0625	166.41	0.0001
AB	2475.0625	1	2475.0625	1.10	0.3252
AC	94,402.5625	1	94,402.5625	41.91	0.0002
BC	18.0625	1	18.0625	0.01	0.9308
ABC	126.5625	1	126.5625	0.06	0.8186
Error	18,020.5000	8	2252.5625		
Total	531,420.9375	15			

ANOVA (Cont.)



(a) The response surface



(b) The contour plot

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_3 x_3 + \hat{\beta}_{13} x_1 x_3$$

$$= 776.0625 + \left(\frac{-101.625}{2}\right) x_1 + \left(\frac{306.125}{2}\right) x_3 + \left(\frac{-153.625}{2}\right) x_1 x_3$$

ANOVA (Cont.)

$$SS_{\text{Model}} = SS_A + SS_B + SS_C + SS_{AB} + SS_{AC} + SS_{BC} + SS_{ABC}$$

$$R^2 = \frac{SS_{\text{Model}}}{SS_{\text{Total}}}$$

$$R^2_{\text{Adj}} = 1 - \frac{SS_E/df_E}{SS_{\text{Total}}/df_{\text{Total}}}$$

Adjusted R^2 tells how much variation is explained **after penalising unnecessary factors.**

$$\boxed{\%\ \text{Contribution of A} = \frac{SS_A}{SS_T} \times 100}$$

$$\boxed{SS_{\text{effect}} = \frac{C^2}{2^k r}}$$

ANOVA (Cont.)

- n = total number of observations (fixed once experiment is done)
- k = number of predictors (factors in the model)

Error degrees of freedom:

$$df_E = n - k - 1$$

Increasing k always reduces $n - k - 1$.
Adjusted R^2 increases only if the new predictor reduces residual variation sufficiently.

%Contribution of the Factors

$$\%A = \frac{41311.56}{531420.94} \times 100 = 7.77\%$$

$$\%B = 0.04\%$$

$$\%C = 70.54\%$$

$$\%AB = 0.47\%$$

$$\%AC = 17.76\%$$

$$\%BC \approx 0.003\%$$

$$\%ABC = 0.02\%$$

$$\%Error = 3.39\%$$

$$R^2 = \frac{SS_{\text{Model}}}{SS_{\text{Total}}} = \frac{5.134 \times 10^5}{5.314 \times 10^5} = 0.9661$$

$$R^2_{\text{Adj}} = 1 - \frac{SS_E/df_E}{SS_{\text{Total}}/df_{\text{Total}}} = 1 - \frac{18{,}020.50/8}{5.314 \times 10^5/15} = 0.9364$$